



Croatian Open Competition in Informatics

Round 5, March 11th 2023

Tasks

Task	Time limit	Memory limit	Score
Kalendar	1 second	512 MiB	50
Diskurs	2 seconds	512 MiB	70
Logaritam	1 second	512 MiB	110
Slastičarnica	2 seconds	512 MiB	110
Zastave	1 second	512 MiB	110
Total			450

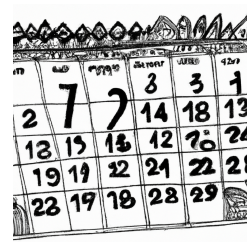


Task Kalendar

Magdalena likes calendars, and she makes her own calendar for each month.

Each day of the month is represented with exactly three characters:

- If the day number is single-digit, then it is represented as `'..X'`. For example, the number 7 is represented as `'..7'`.
- If the day number is double-digit, then it is represented as `'.XY'`. For example, the number 17 is represented as `'.17'`.



Each row of the calendar represents a week, and each week consists of 7 days. If the week doesn't have all the 7 days (because the month doesn't start on Monday, or it doesn't end on Sunday), then the missing days are replaced with `'...'`.

Magdalena also wants her calendar to be pretty. She will decorate it in the following way: she will fill the upper and lower sides with `'-'` (ASCII 45), the left and right sides with `'|'` (ASCII 124), and the corners with `'+'` (ASCII 43).

For example, the format of Magdalena's calendar, when the month has 31 days and starts on Wednesday, is the following:

```
+-----+
|. . . . . 1 . 2 . 3 . 4 . 5 |
|. . 6 . 7 . 8 . 9 . 10 . 11 . 12 |
|. 13 . 14 . 15 . 16 . 17 . 18 . 19 |
|. 20 . 21 . 22 . 23 . 24 . 25 . 26 |
|. 27 . 28 . 29 . 30 . 31 . . . . . |
+-----+
```

Your task is to determine the format of Magdalena's calendar if it has n days, and the first day of the month is the x -th day of the week. For example, if $x = 1$, the month starts on Monday, and if $x = 5$, it starts on Friday.

Note: We assume the first day of the week is Monday.

Input

The first and only line contains integers n and d ($1 \leq n \leq 99, 1 \leq d \leq 7$), the number of days in the month, and the day it starts with.

Output

Print Magdalena's calendar.

Scoring

Subtask	Points	Constraints
1	7	All the days will fit in a row.
2	19	$d = 1$
3	24	No additional constraints.



Examples

input

31 3

output

```
+-----+
|.....1..2..3..4..5|
|..6..7..8..9.10.11.12|
|.13.14.15.16.17.18.19|
|.20.21.22.23.24.25.26|
|.27.28.29.30.31.....|
+-----+
```

input

1 5

output

```
+-----+
|.....1.....|
+-----+
```

input

28 7

output

```
+-----+
|.....1|
|..2..3..4..5..6..7..8|
|..9.10.11.12.13.14.15|
|.16.17.18.19.20.21.22|
|.23.24.25.26.27.28...|
+-----+
```

Clarification of the second example:

Note that, although there is only one day in the month, the calendar still has the format of seven days per row.



Task Diskurs

You are given n non negative integers $a_1, a_2, \dots, a_n$ less than 2^m . For each of them you are to find the maximum possible *hamming distance* between it and some other element of the array a .

The *hamming distance* of two non negative integers is defined as the number of positions in the binary representation of these numbers in which they differ (we add leading zeros if necessary).

Formally, for each i calculate:

$$\max_{1 \leq j \leq n} \text{hamming}(a_i, a_j)$$

Input

The first line contains two integers n and m ($1 \leq n \leq 2^m, 1 \leq m \leq 20$).

The second line contains n numbers a_i ($0 \leq a_i < 2^m$).

Output

Output n numbers separated with spaces, where the i -th number is the maximum *hamming distance* between a_i and some other number in a .

Scoring

Subtask	Points	Constraints
1	20	$m \leq 10$
2	25	$m \leq 16$
3	25	No additional constraints.

Examples

input

4 4
9 12 9 11

output

2 3 2 3

input

4 4
5 7 3 9

output

2 3 2 3

input

4 4
3 4 6 10

output

3 3 2 3

Clarification of the third example:

The numbers 3, 4, 6, 10 can be represented as 0011, 0100, 0110, 1010, in binary. Numbers 3 and 4 differ at 3 places, same as numbers 4 and 10. On the other hand, the number 6 differs in at most 2 places with all other numbers.

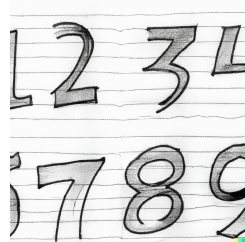
010
100



Task Logaritam

Hrvoje has recently learned about the logarithm function. He really likes the property $\log(xy) = \log(x) + \log(y)$, for each pair of positive real numbers x and y .

He is actually not interested in the function itself, but in logarithmic sequences. A logarithmic sequence of length n is a sequence of real numbers $(a_1, a_2, \dots, a_n)$ for which $a_{xy} = a_x + a_y$ holds for every pair of positive integers x and y such that $xy \leq n$. An example of a logarithmic sequence of length 6 is $0, 1, \pi, 2, 0.7, 1 + \pi$.



For his homework, Hrvoje needed to write q logarithmic sequences of length n , however, after a long night of effort he woke up just to find out that Matej had changed exactly one element of each sequence. Hrvoje doesn't have a lot of time to correct his homework, so he is interested in the least number of elements of each sequence he needs to change so the sequence becomes logarithmic again. Unfortunately, Matej had written his element with a pen, so Hrvoje **cannot** change that element of the sequence.

Hrvoje has forgotten which sequences he wrote for his homework so the only thing he knows is the number of sequences q , the length of each sequence n and the position x_i of the element Matej had changed in the i -th sequence.

Note: It can be proven that for any starting logarithmic sequence the minimal number of changes is the same.

Input

In the first line there are two positive integers n and q ($1 \leq n \leq 10^8$, $1 \leq q \leq 10^4$), the length of each sequence and the number of sequences.

In the i -th of the next q lines there is a positive integer x_i ($1 \leq x_i \leq n$), the index of the element Matej had changed in the i -th sequence.

Output

In the i -th line output -1 if Hrvoje cannot change the other elements of the i -th sequence such that the sequence becomes logarithmic again, otherwise output the minimal number of changes needed to make the sequence logarithmic again.

Scoring

Subtask	Points	Constraints
1	19	$n \leq 20$, $q \leq 20$
2	26	$q \leq 8$
3	29	$n \leq 10^4$
4	36	No additional constraints.



Examples

input	input	input
6 6	20 5	10000 4
1	7	1234
2	8	2345
3	2	3456
4	19	7890
5	12	
6		output
output	output	15
-1	1	148
2	9	3332
1	9	37
2	0	
0	5	
1		

Clarification of the first example:

If the starting sequence was $0, 1, \pi, 2, 0.7, 1 + \pi$ and Matej changes the fourth element to 8, Hrvoje can change the second element to 4 and the sixth to $4 + \pi$, after which the sequence $0, 4, \pi, 8, 0.7, 4 + \pi$ will be logarithmic again.

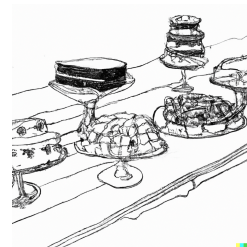


Task Slastičarnica

There is a new bakery in town! Come and try delicious cakes from Dodo's bakery!

The bakers have prepared n cakes, the i -th of which has sweetness a_i . They are displayed on a long table in the given order. There are q people waiting in line to order the best cakes in town. Each of them has an order of the form: "I would like to buy d_i cakes whose sweetnesses are at least s_i ".

Dorijan serves customers in a peculiar way. He will give d_i continuous cakes from the table. To keep the table looking as nice as possible he will only give cakes from the left or the right edge of the table. He noticed that he cannot serve a lot of customers that way, so before serving a customer he will eat some cakes (possibly none) from the edges of the table.



If Dorijan cannot satisfy a customer, he will close the bakery for the day. What is the largest number of customers he can serve, before closing?

Input

The first line contains two integers n, q ($1 \leq n \leq 5000, 1 \leq q \leq 2 \cdot 10^5$), the number of cakes and the number of people in the line.

The second line contains n numbers a_i ($1 \leq a_i \leq 10^9$), where a_i is the sweetness of the i -th cake.

In each of the next q lines there are two integers d_i, s_i ($1 \leq d_i \leq n, 1 \leq s_i \leq 10^9$), the order of the i -th customer in line.

Output

In the first and only line output the number of customers Dorijan can serve.

Scoring

Subtask	Points	Constraints
1	21	$n, q \leq 100$
2	31	$d_1 = d_2 = \dots = d_q = 1$
3	58	No additional constraints.

Examples

input

```
5 6
1 2 3 4 5
1 1
1 2
1 3
1 4
1 6
1 5
```

output

```
4
```

input

```
5 3
1 3 2 2 1
3 1
1 3
2 2
```

output

```
2
```

input

```
9 5
1 3 2 5 1 4 6 2 1
3 2
2 3
1 1
1 2
1 1
```

output

```
4
```



Clarification of the third example:

Dorijan first eats one cake from the left, then gives the next three cakes with sweetnesss 3, 2, and 5 to the first customer. He then eats another cake from the left and gives the next two cakes with sweetnesss 4 and 6 to the second customer. The third customer gets a cake from the right with sweetness 1 and the fourth gets the last cake with sweetness 2. Dorijan unfortunately does not have more cakes, so the last customer goes home empty-handed.



Task Zastave

After an exhausting day of preparing COCI, after sleeping for only three hours and in intervals of 20 minutes, and finally after naughty Patrick and Josip got on his nerves, Vito fell asleep.

Vito was always a pacifist and as a sign of his resignation in front of the disobedience of his (un)reliable friends, Vito dreamed of n white flags. The white flags had the shape of a right triangle swirling in the air with one of their sides parallel to the ground. In the morning, Vito could only remember a few key details... the length of the hypotenuse of the i -th flag was r_i and the total sum of heights of the flags was at most S .



Now awake, he decided he shall fight on the beaches and never surrender! He rushed to the nearest paint shop so that next time he dreams of the n white flags he can paint them over! But he quickly realized, he isn't sure how much paint he has to buy. So he asked you to calculate the maximum possible total area of the n white flags satisfying the constraints!

Input

The first line contains the integers n and S ($1 \leq n \leq 100\,000, 1 \leq S \leq 10^{10}$), the number of flags and the maximum possible sum of heights of the flags.

In the next line there are n integers r_i ($1 \leq r_i \leq 100\,000$).

Output

In the only line, output the maximum possible sum of areas of the flags. Your solution will be considered correct if the absolute or relative error is smaller than 10^{-6} .

Scoring

Subtask	Constraints	Restrictions
1	41	$n \leq 100$
2	22	$n \leq 1000$
3	47	No additional constraints.

Examples

input

2 3

4 5

output

6.5200982141

input

1 6

10

output

24.0000000000

input

4 7

5 5 6 6

output

18.5706715170

Clarification of the second example

The largest possible area is achieved by a flag with sides 6, 8, and 10 and the total area is 24.